

Govt. Graduate College for Women, Baghbanpura, Lahore
BS Physics IV semester

Philosophy of Science

Short Questions

Q1. What is the simple view of science?

Ans. The simple view of science is that we tend to make large number of observations about any aspect of nature and on the bases of these unprejudiced and objective observations, we make general or universal conclusions.

Q2. How did Frances Bacon describe the method of science? /What is inductive method?

Ans. Francis Bacon introduced humanity to induction as the most reliable scientific method for gaining verifiable knowledge. According to this method we take a single example of an aspect of nature and then replicate the observations as many times as we can and with varying conditions. If all observations, thus made, give us the same result we can generalize these results into a law or rule. For example we observed that

- A pen falls to the ground
- A ball falls to the ground
- A feather falls to the ground

Therefore all things fall to the ground.

Q3. How philosophy helps in the progress of science?

Ans. Philosophy helps science to progress by giving it rules and principles to work in a systematic way. Philosophy provides science with new problems, news ideas and modes of inquiry. It is rightly said that where science ends, philosophy begins.

Q4. what is the problem of induction?

Ans. Scientific method works on the principle of induction. However induction has a major flaw in it that it tends to make bold predictions about things on the basis of a few examples. Science believes that things will continue to happen in the future in the same way as they have happened in the past. This is known as law of causation where the same set of causes have the same set of effects.

In reality there seems to be no logical necessity between cause and effect and so there is no certainty that things will happen in the future as they have happened in the past. It was David Hume who had criticized induction on the basis of law of causation.

Q5. What is meant by inductive leap?

Ans. Inductive leap is a tool used in induction where from the results obtained from some observations about some natural phenomena, we make a general or universal or comprehensive conclusion. This method is particularly helpful in instances where either the time period required to make a single observation is too large or the number of possible observations is too large. Whatever the case may be the fact is that we can never reach certainty.

Q6. What is generalization?

Ans. Generalization is the process of making universal conclusions on the basis of particular observations. For example 'All crows are black'. On the basis of nature of observation, generalization is usually of two types

1. Empirical generalization; where only observation is necessary, no need for experiment. For example 'All mangoes are sweet'
2. Scientific generalization; where we need to find the causal relation between matter at hand and so require experimentation. For example 'B.P of all liquids increase with the increase in pressure.

Q7. What is law of causation?

Ans. Law of causation states that nothing occurs in the world coincidentally. Events are always related to each other in a set pattern where one event is considered as cause and the one immediately following is known as effect. For example dark Grey clouds in the sky is the cause and rain is the effect.

Q8. What is law of uniformity of nature?

Ans. The law of uniformity of nature states that nature always follows a course. Every natural phenomenon, everything we observe has been happening in a pattern in the past and will follow the same pattern in the future. For example all things fall to the ground, have been falling to the ground in the past and will continue to fall to the ground in the future.

Q9. How did Hume criticized the process of induction?

Ans. Hume criticized the process of induction by rejecting the law causation. According to the law events in nature are not isolated rather a combination cause and effect. This law has been put to use while making all of scientific laws. However, according to Hume, the events arranged as cause and effect are not logically connected and there is no certainty and necessity that this will continue to do so in the future.

Q10. What is Falsificationism?

Ans. Falsificationism is a system of scientific method proposed by Karl Popper. According to the theory, instead of trying to prove the validity of scientific information, a scientist should try to look for example that disapproves a theory. The theory that has fewer instances of being false is more credible.

11. How does principle of falsifiability works?

Ans. In its basis, falsifiability is a principle that for any hypothesis to have credibility, it must be inherently disprovable before it can become accepted as a scientific hypothesis or theory. It is a more appropriate way to discard pseudo-sciences.

12. What is the main objection to falsifiability?

Ans. The main objection to falsificationism is that it does not give importance to the role of confirmation of hypothesis in science. For example if I make a hypothesis that

boiling point of water varies at a constant rate as the pressure of atmosphere varies, I can successfully predict about a number of observations on the lines of this hypothesis and I can also make an accurate prediction that mountaineers would not be able to make a good cup of tea because of high altitude. Although it is an accurate prediction but it is neglected in falsificationism.

13. Define Logical Positivism?

Ans. Logical positivism is a philosophical approach that emphasizes the importance of empirical evidence in determining the truth of propositions. This concept was introduced at the height of rapid scientific progress. Its aim was to filter out pseudo sciences from actual scientific endeavours. It rejects metaphysics and idealistic claims and asserts that knowledge can only be derived from scientifically confirmed statements.

14. Define Paradigm shift?

Ans. A paradigm shift is a standard, perspective or a set of ideas. It is a way of looking at things. A paradigm shift occurs when scientist tend to get more dead ends in their scientific observations and experiments. When scientific activity and experimentation begin to contradict premises the experts previously considered unshakable, a new set of ideas, standards and laws are put in force replacing the old ones.

15. Define Epistemological anarchism?

Ans. Epistemological anarchism is an epistemological theory that holds that there are no useful or exception free methodological rules governing the progress of science and growth of knowledge.